Task 3.1:

Source Code:

```cpp
#include <iostream>
using namespace std;

int main()
{
    // Declaring Variables
    int a;
    float b;
    char c;

    //Taking Inputs from user

    cout<<"Enter an Integer value:\n";
    cin>>a;

    cout<<"Enter a floating value:\n";
    cin>>b;

    cout<<"Enter a character:\n";
    cin>>c;

    //Displaying the output

    cout<<"Your entered integer is: "<<a<<"\n";
    cout<<"Your entered float is: "<<b<<"\n";
    cout<<"Your entered character is: "<<c<<"\n";


    return 0;
}
```

Output:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
rehanshahid@Muhammads-MacBook-Air 06:10:26 % g++ task3_1.
rehanshahid@Muhammads-MacBook-Air 06:10:26 % ./task3_1
Enter an Integer value:
876
Enter a floating value:
67.786
Enter a character:
r
Your entered integer is: 876
Your entered float is: 67.786
Your entered character is: r
rehanshahid@Muhammads-MacBook-Air 06:10:26 %
```

Task 3.2:

Source code:

task3_2.cpp > main()

```cpp
#include <iostream>
//#include <windows.h> I cannot include windows.h in my program because i am using mac. i will install windows in vm then i will practice it
#include <cstdlib> // library to import system sound from mac
using namespace std;

int main()
{
    //Declaring variables
    int a,b;

    //Taking Inputs from user
    cout<<"Enter an integer a:\n";
    cin>>a;
    cout<<"Enter an integer b:\n";
    cin>>b;

    //Displaying results
    // Beep(700,2000);  Beep function of windows.h
    cout<<"Sum:\t\t"<<a<<" + "<<b<<" = "<<a+b<<"\n";
    cout<<"Difference:\t"<<a<<" - "<<b<<" = "<<a-b<<"\n";
    cout<<"Product:\t"<<a<<" * "<<b<<" = "<<a*b<<"\n";
    cout<<"Quotient:\t"<<a<<" / "<<b<<" = "<<a/b<<"\n";
    cout<<"Remainder:\t"<<a<<" % "<<b<<" = "<<a%b<<"\n";
    system("afplay /System/Library/Sounds/Ping.aiff");


    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

rehanshahid@Muhammads-MacBook-Air 06:10:26 % g++ task3_2.cpp -o task3_2
rehanshahid@Muhammads-MacBook-Air 06:10:26 % ./task3_2
Enter an integer a:
50
Enter an integer b:
26
Sum:            50 + 26 = 76
Difference:     50 - 26 = 24
Product:        50 * 26 = 1300
Quotient:       50 / 26 = 1
Remainder:      50 % 26 = 24
rehanshahid@Muhammads-MacBook-Air 06:10:26 %
```

Task 3.3

Source code:

```cpp
task3_3.cpp > main()
#include <iostream>
#include <cstdlib> // Library to import system sound from mac
// #include <windows.h>   I cannot include windows.h in my program because i am using mac. i will install windows in vm then i will practice it
using namespace std;

int main()
{
    // Printing MCS and beep from system call
    cout<<"MCS\n";
    system("afplay /System/Library/Sounds/Ping.aiff"); //system funtions take arg and run it directly on terminal, it fetches and then give us sound from our library

    cout<<"\tMCS\n";
    system("afplay /System/Library/Sounds/Ping.aiff");

    cout<<"\t\tMCS\n";
    system("afplay /System/Library/Sounds/Ping.aiff");

    cout<<"\t\t\tMCS\n";
    system("afplay /System/Library/Sounds/Ping.aiff");

    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
rehanshahid@Muhammads-MacBook-Air 06:10:26 % ./task3_3
MCS
        MCS
                MCS
                        MCS
rehanshahid@Muhammads-MacBook-Air 06:10:26 %
```

Taks 3.4

Source code:

```cpp
task3_4.cpp > main()
#include <iostream>
#include <cstdlib> // Library to import system sound from mac
// #include <windows.h>   I cannot include windows.h in my program because i am using mac. i will install windows in vm then i will practice it
using namespace std;

int main()
{

    // Displaying the given information
    cout<<"This is a test.\n";
    cout<<"He asked, \"How are you doing?\"\n";
    cout<<"\"Education is the most powerful weapon which you can use to \'CHANGE\' the world.\n";
    // 5 Beeps
    system("afplay /System/Library/Sounds/Ping.aiff"); // system funtions take arg and run it directly on terminal, it fetches and then give us sound from our library.
    system("afplay /System/Library/Sounds/Ping.aiff");
    system("afplay /System/Library/Sounds/Ping.aiff");
    system("afplay /System/Library/Sounds/Ping.aiff");
    system("afplay /System/Library/Sounds/Ping.aiff");

    return 0;
}
```

Output

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

rehanshahid@Muhammads-MacBook-Air 06:10:26 % g++ task3_4.cpp -o task3_4
rehanshahid@Muhammads-MacBook-Air 06:10:26 % ./task3_4
This is a test.
He asked, "How are you doing?"
"Education is the most powerful weapon which you can use to 'CHANGE' the world.
rehanshahid@Muhammads-MacBook-Air 06:10:26 %
```

Post-Lab Task:

Source code:

task3_pl.cpp > main()

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{

    // Declaring variable
    float n;
    int powR;

    // Taking Output
    cout<<"Enter a number on which you want various nathematical operations:\n";
    cin>>n;

    cout<<"Enter a power you want to raise your number:\n";
    cin>>powR;

    //Displaying data
    cout<<"\nSin of "<<n<<" is:\t\t\t"<<sin(n)<<"\n";

    cout<<"Cos of "<<n<<" is:\t\t\t"<<cos(n)<<"\n";

    cout<<"Tan of "<<n<<" is:\t\t\t"<<tan(n)<<"\n";

    cout<<"Natural log of "<<n<<" is:\t\t"<<log(n)<<"\n";

    cout<<"Common log of "<<n<<" is:\t\t"<<log10(n)<<"\n";

    cout<<"Exp of "<<n<<" is:\t\t\t"<<exp(n)<<"\n";

    cout<<n<<" raised to power "<<powR<<" is:\t"<<pow(n,powR)<<"\n";

    cout<<"Sqrt of "<<n<<" is:\t\t\t"<<sqrt(n)<<"\n";

    cout<<"Ceiling of "<<n<<" is:\t\t"<<ceil(n)<<"\n";

    cout<<"Floor of "<<n<<" is:\t\t"<<floor(n)<<"\n";


    return 0;
}
```

Output:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

rehanshahid@Muhammads-MacBook-Air 06:10:26 % g++ task3_pl.cpp -o task3_pl
rehanshahid@Muhammads-MacBook-Air 06:10:26 % ./task3_pl
Enter a number on which you want various nathematical operations:
5.4
Enter a power you want to raise your number:
3

Sin of 5.4 is:                  -0.772764
Cos of 5.4 is:                  0.634693
Tan of 5.4 is:                  -1.21754
Natural log of 5.4 is:          1.6864
Common log of 5.4 is:           0.732394
Exp of 5.4 is:                  221.406
5.4 raised to power 3 is:       157.464
Sqrt of 5.4 is:                 2.32379
Ceiling of 5.4 is:              6
Floor of 5.4 is:                5
rehanshahid@Muhammads-MacBook-Air 06:10:26 %
```